

CSE-2430 Mid Examination Spring 2025

Preliminary Test

D1

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 32 MB

Ashik is going to attend an exam to become an army officer, where he needs to pass the preliminary exam first. He is given a question where he is given an array B of N numbers, he has to find the minimum result of $B_j + B_i + i - j \leq N$, where $1 \leq j < i \leq N$, so that he get the chance to the next step

Input

The first line contains a number T ($1 \leq T \leq 100$) number of test cases.

Each test case contains two lines:

- The first line consists a number N ($2 \leq N \leq 100$) number of elements.
- The second line contains N numbers ($-10^6 \leq B_j \leq 10^6$).

Output

For each test case print a single line contains **the smallest** possible sum for the corresponding test case

Sample Input

1

4

20 1 9 4

Sample Output

7

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D2

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 32 MB

Upside Down Game

Jubayer is facing a problem to solve an IQ question, where he is given a string S . He needs to print the **origin** string after replacing the following:

- Replace every **comma** character ',' with a ' ? ' character.
- Replace every **capital character** in S with its respective **small character** and **Vice Versa**.

Input

Only one line contains a string S ($1 \leq |S| \leq 10^5$) where $|S|$ is the length of the string and it consists of **lower** and **upper** English letters and **comma** character ','.

Output

Print the string after the **conversion**..

Sample Input

```
happy,NewYear,enjoy
```

Sample Output

```
HAPPY ? nEWyEAR ? ENJOY
```

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D3

Count Distinct Numbers

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 32 MB

You are given N numbers, where you have to find the maximum distinct numbers from that N numbers

Input

Each test case contains two lines:

- The first line consists of a number N ($2 \leq N \leq 100000$) number of elements.
- The second line contains N numbers ($1 \leq B_j \leq 10^{18}$).

Output

Print the total distinct numbers.

Sample Input

```
5
10 10 10 5 6
```

Sample Output

```
3
```